SafeHer — 13-Slide Speaker Script & Viva Guide

BMS Institute of Technology & Management · Department of Master of Computer Applications
Project Work Phase I (MMCA34) · Guide: Dr. M Sridevi · Lead Presenter: Anusha K (1BY25MC017)

Instructions for Anusha: This document contains the exact spoken lines for each of your 13 PowerPoint slides, timed for a confident 10-minute presentation. Each slide also includes potential examiner questions from Dr. M Sridevi / the review panel and critical guidelines on what to emphasize and what to avoid.

SLIDE 01 OF 13

Target Duration: **45 seconds**

Visual: *Navy/Gold presentation cover displaying college name, SafeHer logo, Complete Project Vision, Module 1 (Anusha K), Module 2 (Aman Yadav), Guide Dr. M Sridevi, and SDG 5 Target 5.2.*

Title Slide

WHAT TO SAY (Verbatim Spoken Script):

"Good morning, ma'am and respected evaluators.

We are from the Department of MCA, BMS Institute of Technology and Management.

Our complete project is titled SafeHer: an integrated data-driven platform for women's safety in public spaces.

SafeHer bridges crowdsourced citizen incident reporting with backend geospatial analysis, hotspot identification, and risk assessment to provide authorities with proactive, preventive patrol intelligence rather than delayed reactions.

Our project is guided by Dr. M Sridevi.

It directly aligns with United Nations Sustainable Development Goal 5 – Gender Equality, specifically Target 5.2, which focuses on eliminating all forms of violence against women and girls in public spaces.

To build this end-to-end system comprehensively, our project is divided into two complementary, interconnected modules:

Module 1 is my module: "Security provision through hotspot identification and risk assessment for women's safety in public spaces."
I am Anusha K, 1BY25MC017. My individual scope is the analytical core—handling data cleaning, spatial clustering on a 1 km grid, hotspot detection, multi-tier risk stratification, and analytical APIs.

The second module is Module 2: "Women's safety and dynamic security monitoring mobile application," which is handled by Aman Yadav, 1BY25MC012. His scope covers citizen mobile incident reporting and the authority administrative dashboard.

Both modules communicate seamlessly through REST APIs to deliver the unified SafeHer platform."

IF EXAMINER / GUIDE ASKS:

If asked what SafeHer represents: SafeHer is the complete working system uniting citizen crowdsourced reporting with backend geospatial intelligence. The official registered examination titles are the two approved module titles.

WHAT TO AVOID / CRITICAL CAUTION:

Do not jump straight into module divisions before introducing the complete SafeHer project vision, guide Dr. M Sridevi, and SDG 5 context.

SLIDE 02 OF 13

Target Duration: **50 seconds**

Visual: *Three-pillar comparison: Existing Difficulties (fragmented registers) vs SafeHer Data-Driven Approach (4 pillars) vs Actionable Authority Outcomes.*

Introduction & Project Context

WHAT TO SAY (Verbatim Spoken Script):

"The main objective of our project is to improve women's safety in public spaces using a data-driven approach.

In the existing system, incidents may be reported through different sources, but it can be difficult to identify where incidents are repeatedly occurring and which locations require more attention.

Our project combines incident reporting, location-based analysis, hotspot identification and risk assessment.

The mobile application allows users to report safety-related incidents by providing details such as the incident type, date, time, description and geographical location.

The reported information is then processed and analysed.

Module 1 focuses on analysing the incidents and identifying hotspots and risk levels, while Module 2 focuses on the mobile application and authority dashboard.

The main purpose is to provide useful information for authorities so that they can take preventive actions such as better monitoring and patrolling in repeatedly affected areas."

IF EXAMINER / GUIDE ASKS:

If asked how this differs from police registers or 112 calls: Traditional calls and registers record incidents in isolation. SafeHer identifies spatial recurrence—grouping multiple reports onto specific 1 km street grids so authorities know exactly where repeated incidents happen.

WHAT TO AVOID / CRITICAL CAUTION:

Do not claim we are replacing the police. We are providing actionable spatial intelligence to assist authorities in patrol allocation.

SLIDE 03 OF 13

Target Duration: **55 seconds**

Literature Survey & Research Foundation

Visual: *Four foundational pillars: Sherman (1989) Hotspot Theory, Chainey & Ratcliffe (2005) GIS Crime Mapping, Environmental Determinants (Nehme et al., 2017 & Safetipin Audits), and Research Gap.*

WHAT TO SAY (Verbatim Spoken Script):

“Now I will present the literature survey and theoretical foundation of our project.

Our work is grounded on well-established research in GIS crime mapping and environmental criminology.

First, research by Sherman in 1989 on hotspot theory proved that crime is not distributed evenly across a city, but concentrates heavily in specific micro-places. This justifies why hotspot identification is a scientifically valid planning tool.

Second, Chainey and Ratcliffe in 2005 demonstrated that GIS-based hotspot detection allows law enforcement to allocate patrolling and civic resources efficiently. In SafeHer, we use a 1-kilometer spatial grid to evaluate spatial density and recurrence.

Third, environmental studies, such as Nehme et al. in 2017 and Safetipin audit studies, show that physical infrastructure like street lighting, CCTV surveillance, and police proximity strongly influence women’s perceived and actual safety.

Finally, the research gap we address is that existing mobile applications are mostly reactive SOS panic buttons, while official police records are delayed and coarse. SafeHer bridges this gap by providing a proactive, rule-based, street-level geospatial hotspot and risk assessment platform using crowdsourced community data.”

IF EXAMINER / GUIDE ASKS:

If asked what literature foundation supports SafeHer: 'We rely on Sherman (1989) establishing that crime concentrates at micro-places, Chainey & Ratcliffe (2005) demonstrating GIS grid hotspot allocation for police patrols, and environmental audit studies showing lighting and surveillance govern public safety perception.'

WHAT TO AVOID / CRITICAL CAUTION:

Do not claim you invented hotspot theory from scratch. Acknowledge established criminological and GIS research.

SLIDE 04 OF 13

Target Duration: **60 seconds**

Datasets and Resources

Visual: *Structured data architecture showing Incident Data (types, timestamps, GPS), Bengaluru Street Safety Dataset (roads, lighting, surveillance), and OpenStreetMap integration.*

WHAT TO SAY (Verbatim Spoken Script):

“For our project, we use different types of data and resources.

The first important source is the incident data. This contains information related to reported incidents, including their location, date, time and type.

We also use geographical and public infrastructure information.

For geographical information, we use OpenStreetMap and related geographical resources.

The available Excel dataset is currently focused on Bengaluru.

The dataset contains information related to roads, streets and buildings, along with prototype or assumed infrastructure attributes such as lighting and surveillance.

These infrastructure attributes are used for the project analysis and demonstration.

In addition to the existing dataset, new incident information can come from the mobile application.

This allows the system to combine existing data with newly reported incidents."

IF EXAMINER / GUIDE ASKS:

If asked why Bengaluru is used and what is in the Excel file: The dataset 'bengaluru_street_safety_REAL.xlsx' provides structured street networks and physical safety attributes (streetlights, CCTV presence, bus stops) to validate our hotspot and risk scoring models. The system architecture is built to scale across other cities and Karnataka districts.

WHAT TO AVOID / CRITICAL CAUTION:

Do not say infrastructure attributes are random. State they are prototype/modelled attributes from OpenStreetMap and survey proxies.

SLIDE 05 OF 13

Target Duration: **60 seconds**

Visual: *Full end-to-end data pipeline diagram from citizen mobile report -> REST API -> Database -> Preprocessing -> Spatial Hotspots -> Risk Tiers -> Authority Dashboard.*

Complete System Data Flow

WHAT TO SAY (Verbatim Spoken Script):

"This slide shows the complete data flow of our project.

First, we have the existing data and the new incident data collected through the mobile application.

The incident information is sent through an API to the backend.

The backend stores the information in the database.

After that, Module 1 performs data preprocessing and incident analysis.

The data is analysed based on factors such as incident type, date, time, location, frequency and recurrence.

Then we perform geospatial analysis to understand where incidents are concentrated.

Based on this analysis, we identify hotspots and assign different risk levels such as Low, Medium and High.

The analytical results are then provided through APIs to the Authority Dashboard.

The dashboard allows authorities to view the information using maps, charts and filters.

The system can also support alerts and preventive patrolling based on repeated or concentrated historical incidents."

IF EXAMINER / GUIDE ASKS:

If asked whether the mobile app computes hotspots: No. The mobile app only sends raw incident coordinates. All heavy spatial processing, grid snapping, and risk assignment happen in Module 1 on the backend.

WHAT TO AVOID / CRITICAL CAUTION:

Do not rush this slide. Trace the arrows left-to-right from data capture to delivery.

SLIDE 06 OF 13

Target Duration: **50 seconds**

Visual: *System architecture diagram paired with categorized tech stack: Flutter/React Native, Python/FastAPI, MongoDB, Pandas/GeoPandas, Leaflet/OSM, React/Next.js.*

Architecture and Tech Stack

WHAT TO SAY (Verbatim Spoken Script):

"This slide shows the architecture and technologies used in our project.

For the mobile application, we can use a cross-platform framework such as Flutter or React Native.

For the backend, we use Python and FastAPI.

The database layer uses MongoDB for storing incident information.

For data processing and analysis, we use libraries such as Pandas, NumPy and GeoPandas.

For geographical visualization, we use OpenStreetMap along with Leaflet or Folium.

For charts and analysis visualizations, we use Matplotlib or Plotly.

The authority dashboard can be developed using technologies such as HTML, CSS, JavaScript and React.

The communication between different components is handled using REST APIs and JSON.

We can use tools such as Postman for API testing, and development tools such as VS Code and Android Studio."

IF EXAMINER / GUIDE ASKS:

If asked why FastAPI was chosen over Flask/Django: FastAPI provides native async support, automatic OpenAPI/Swagger documentation, high performance, and typed request validation with Pydantic.

WHAT TO AVOID / CRITICAL CAUTION:

Do not claim technologies you haven't reviewed. The stack is clean: Python backend, React frontend/mobile, MongoDB database.

SLIDE 07 OF 13

Target Duration: **60 seconds**

Visual: *Side-by-side modular division cards showing Module 1 (Anusha K - Analytical Engine) and Module 2 (Aman Yadav - Client Applications & Dashboard) working as a single unified system.*

Project Modularization & Scope Division

WHAT TO SAY (Verbatim Spoken Script):

"Our project is divided into two major modules, but both modules work together as a single system.

Module 1 is my module.

It focuses on incident analysis, geospatial analysis, hotspot identification and risk-level assessment.

My module takes the available incident and geographical data, processes it, identifies areas where incidents are concentrated or repeatedly reported, and produces the analytical results.

Module 2 is handled by Aman.

It focuses on the women's safety mobile application and the authority dashboard.

The mobile application allows users to submit incident reports.

The authority dashboard allows authorized personnel to monitor the reported incidents, view maps and charts, apply filters and see hotspot and risk information.

So, Module 1 provides the analytical results, while Module 2 provides the reporting and monitoring interface."

IF EXAMINER / GUIDE ASKS:

If asked who does what if Aman is absent: State clearly: 'Module 1 is my complete ownership: data integration, 0.01° grid clustering, risk algorithms, and REST APIs. Aman handles the mobile client and authority UI. Together they form the SafeHer platform.'

WHAT TO AVOID / CRITICAL CAUTION:

Never say 'we both did everything'. Examiners require distinct individual ownership for Phase I evaluation.

SLIDE 08 OF 13

Target Duration: **75 seconds**

Visual: *Detailed 8-stage analytical methodology diagram: Data Integration -> Pre-processing -> Incident Analysis -> Geospatial Binning -> Hotspot Detection -> Risk Stratification -> Alert Generation -> Serving Layer.*

Module 1 Methodology

WHAT TO SAY (Verbatim Spoken Script):

"Now I will explain my module in more detail through its eight-stage analytical pipeline.

The first step is data integration, where we combine geographical context with incident reports received from the mobile application.

Second is data preprocessing, handling missing values, deduplication, and validating GPS coordinates within Karnataka.

Third is incident analysis, categorizing event types, temporal patterns across time of day and week, and recurrence.

Fourth is geospatial binning, mapping citizen reports onto a uniform 1-kilometer spatial grid.

From Step 4, the workflow flows directly into Step 5: hotspot detection, flagging areas with 3 or more incidents across distinct dates.

Sixth is risk stratification, assigning Low, Medium, or High risk tiers using rule-based criteria fusing incident density with infrastructure vulnerabilities.

Seventh is alert generation, ranking high-risk areas to produce a prioritized preventive patrol list for law enforcement.

Finally, eighth is the serving layer, publishing GeoJSON and REST APIs so that analytical results seamlessly feed the Authority Dashboard."

IF EXAMINER / GUIDE ASKS:

If asked for exact thresholds: Spatial snapping uses a 0.01° grid cell (~1 km x 1 km). A cell with >= 3 incident reports is a Hotspot. High Risk: >= 8 reports, or >= 5 reports with >= 40% harassment severity. Medium Risk: >= 4 reports. Low Risk: remaining hotspots.

WHAT TO AVOID / CRITICAL CAUTION:

CRITICAL: Do NOT claim machine learning prediction or predictive policing. Emphasize: 'Descriptive spatial clustering of historical incidents.'

SLIDE 09 OF 13

Target Duration: **60 seconds**

Visual: *Visual output display: Hotspots map with color-coded risk markers (Red=High, Amber=Med, Green=Low) alongside statistical breakdown cards (72 Hotspots, 30 High, 19 Med, 23 Low, 55 Alerts).*

Module 1 Sample Analytical Results

WHAT TO SAY (Verbatim Spoken Script):

"This slide shows the type of output produced by Module 1.

The purpose of this output is to make the incident analysis easier to understand visually.

Instead of looking at individual incident records, authorities can see where incidents are concentrated.

The hotspot information can be represented on a map, along with the corresponding risk level.

The system can categorize identified areas into Low, Medium and High risk levels.

This makes it easier for authorities to focus their attention on areas with repeated or concentrated incidents.

The results can also support preventive patrolling and security planning.

So, the main output of Module 1 is not just the raw incident data, but a more useful representation of the data through statistics, geographical information, hotspots and risk levels."

IF EXAMINER / GUIDE ASKS:

If asked what the numbers mean: In our sample validation run of 1,079 reports, 534 were incident-like, forming 72 active hotspot cells. 30 are High risk requiring immediate patrol focus, 19 are Medium risk, and 23 are Low risk, producing 55 daily preventive patrol alerts.

WHAT TO AVOID / CRITICAL CAUTION:

Do not claim these are finalized government FIR statistics. State: 'This is the verified sample demonstration run proving the pipeline method.'

SLIDE 10 OF 13

Target Duration: **60 seconds**

Visual: *Authority Dashboard screenshot and architectural cards for Citizen Mobile App, Dynamic Security Monitoring, and Multi-Factor Authority Filters.*

Module 2: Mobile App & Authority Dashboard

WHAT TO SAY (Verbatim Spoken Script):

"Now I will explain Module 2.

Module 2 focuses on the Women's Safety and Dynamic Security Monitoring Mobile Application.

The application allows users to report women-safety-related incidents.

While submitting a report, the user can provide information such as the incident type, date, time, description and geographical location.

The information is sent through the API to the backend and stored in the database.

The important feature is that new reports can continuously enter the system.

This is why the system supports dynamic security monitoring.

The new incident information can then be processed by Module 1.

The Authority Dashboard provides authorized personnel with a centralized view of the information.

Authorities can view incident statistics, geographical information, hotspots and risk levels.

They can also filter the information based on factors such as region or city, incident type, date, time and risk level.

The system can also generate daily alerts based on recent or recurring incidents to support preventive patrolling."

IF EXAMINER / GUIDE ASKS:

If asked why dynamic monitoring is important: Static crime maps become outdated immediately. By allowing continuous crowdsourced mobile reporting, the map dynamically reflects emerging safety concerns in real time.

WHAT TO AVOID / CRITICAL CAUTION:

Do not claim live video streaming or facial recognition. The system captures citizen safety feedback, incident logs, and GPS.

SLIDE 11 OF 13

Target Duration: **60 seconds**

SafeHer System Interfaces & UX

Visual: *Showcase of 3 system UI screens: Mobile Report Screen (incident types, GPS), Citizen Community Map (safe/unsafe pins), and Authority Command Dashboard with multi-parameter filters.*

WHAT TO SAY (Verbatim Spoken Script):

"This slide shows the main screens or interfaces of our proposed system.

The first part is the mobile application.

The user can access the application and submit an incident report by providing the required information and geographical location.

The second part is the Authority Dashboard.

The dashboard provides a centralized interface for authorized personnel.

It can display incident information using maps, charts and statistics.

The authority can apply filters to focus on a particular region, incident type, date, time or risk level.

The hotspot and risk information generated by Module 1 can also be visualized on the dashboard.

This makes the system easier to use because the authority does not have to manually analyse individual incident records."

IF EXAMINER / GUIDE ASKS:

If asked about user privacy: Citizen reports are anonymous by design. Personal identity, names, and contact numbers are not required to submit a report, preventing privacy leaks while empowering participation.

WHAT TO AVOID / CRITICAL CAUTION:

Do not confuse the citizen screen with the officer dashboard. The mobile app is for citizens; the dashboard with filters and patrol alerts is strictly for authorized personnel.

SLIDE 12 OF 13

Target Duration: **60 seconds**

Conclusion & Project Contributions

Visual: *Four synthesis cards: Unified Platform, Preventive Patrolling, Bengaluru Validation & Scalability, and SDG 5 Target 5.2 Alignment.*

WHAT TO SAY (Verbatim Spoken Script):

"To conclude, our project combines women-safety incident reporting with data analysis and security monitoring.

Module 1 focuses on analysing historical incidents, performing geospatial analysis, identifying hotspots and assigning risk levels.

Module 2 provides the mobile application and Authority Dashboard through which incidents can be reported and monitored.

The two modules are connected through the backend and APIs.

The main benefit of the system is that it converts incident data into useful information such as hotspots, risk levels,

geographical visualizations and alerts.

This information can help authorities understand areas where incidents are repeatedly or heavily concentrated and support preventive patrolling and safety planning.

Our current available dataset is focused on Bengaluru, while the overall system architecture can be extended to other regions or cities when suitable data is available.

The project contributes to SDG 5 – Gender Equality, especially by supporting efforts to address violence and improve safety for women in public spaces.

Thank you."

IF EXAMINER / GUIDE ASKS:

If asked about future work for Phase II: In Phase II, we will integrate live authenticated field deployment with local police commissionerate feeds and expand street coverage across additional Karnataka municipal corporations.

WHAT TO AVOID / CRITICAL CAUTION:

Do not make up speculative features. Stick to the robust 4 pillars: unified platform, prevention, scalability, and SDG 5.

SLIDE 13 OF 13

Target Duration: **15 seconds**

Thank You & Viva Defense

Visual: *Navy/Gold closing slide displaying SafeHer, BMSIT&M; Department of MCA, Guide Dr. M Sridevi, Anusha K (1BY25MC017), and Aman Yadav (1BY25MC012).*

WHAT TO SAY (Verbatim Spoken Script):

"Thank you, ma'am.

I am ready to answer any questions."

IF EXAMINER / GUIDE ASKS:

Maintain composure, wait for the examiner to finish speaking before answering, and refer back to your slide deck if needed.

WHAT TO AVOID / CRITICAL CAUTION:

Do not rush or speak nervously after saying thank you. Stand confidently and listen to the guide's feedback.